

Group work problem solving session 1

September 14, 2026

There are three exercises below ordered by the difficulty level. We recommend you to attempt these questions following the order. No AI is allowed.

Problem 1: Product mix in a furniture workshop

A small workshop makes chairs and tables. Each chair contributes a profit of \$30 and each table a profit of \$50. Production requires time in two departments, carpentry and finishing, according to the table below. The workshop has 40 carpentry hours and 12 finishing hours available this week.

	Chair	Table	Hours available
Carpentry (hours per unit)	2	4	40
Finishing (hours per unit)	1	1	12
Profit per unit	\$30	\$50	

Fractional units are acceptable, since unfinished items may be carried over to next week.

Tasks.

1. Formulate a linear program that maximises weekly profit.
2. Solve the LP, by any method you can think of.

If you solved Problem 1 easily, move on to Problem 2(b). If you struggled and wish to see another simple example like Problem 1, try Problem 2(a).

Problem 2 (a): Blending animal feed

A mill must produce a 100 kg batch of animal feed by blending corn, soybean meal and wheat bran. The composition and prices are:

	Corn	Soybean meal	Wheat bran
Cost (\$ per kg)	0.30	0.90	0.35
Protein (% by weight)	9	60	20
Fibre (% by weight)	2	6	8

The finished blend must contain **at least 30% protein** and **at most 5% fibre**, both measured as percentages of the weight of the blend.

Tasks.

1. Formulate the problem of minimising the cost of the batch as a linear program.
2. Can you still solve this LP using the method you used in solving Problem 1? (Briefly describe how you would solve the LP if your answer is yes, without doing any calculations.)

Problem 2 (b): Multi-period model

A factory plans production over three months. The demand, unit production cost, and production capacity in each month are shown below.

Month t	Demand d_t (units)	Production cost c_t (dollars per unit)	Production capacity (units)
1	40	4	60
2	70	6	60
3	50	5	60

The factory has 10 units in inventory at the beginning of month 1. Units produced in one month may be stored and used to meet demand in a later month. Holding one unit in inventory at the end of a month costs \$0.50. Shortages and back orders are not allowed. In addition, the factory must have at least 5 units in inventory at the end of month 3. The production and inventory quantities may be fractional.

Tasks. Formulate a linear program that minimizes total production and inventory-holding cost over the three months. (*hint*: introduce variable I_t for the inventory remaining at the end of month t .)

Problem 3: Staffing with a five-day working week

A service desk operates seven days a week and needs at least the following number of staff on duty each day:

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Staff required	18	16	15	16	19	14	12

Every employee works **five consecutive days** and then has **two consecutive days off**, and this pattern repeats every week. The schedule must therefore be the same every week. Management wants to know the smallest number of employees that can staff the desk.

Tasks.

1. Explain why “the number of employees on duty on day j ” is a poor choice of decision variable, and propose a better one.
2. Formulate the problem as a linear program.
3. Is there anything concerning in your model? Is it guaranteed that the optimal solution of your LP must be a valid solution to this real-world problem?

Solutions

- **Problems 1 and 2** require only that we name the obvious quantities: the decision variables are handed to us by the statement of the problem.
- **Problem 3** has decision variables that are not the quantities the problem talks about. Finding the right variables is the entire difficulty; once they are chosen properly, the constraints are easy.

Solution to Problem 1

Formulation

Let

x = number of chairs made this week, y = number of tables made this week.

The linear program is

$$\begin{array}{ll} \text{maximise} & 30x + 50y \\ \text{subject to} & 2x + 4y \leq 40 \quad (\text{carpentry}) \\ & x + y \leq 12 \quad (\text{finishing}) \\ & x, y \geq 0. \end{array}$$

Solving the LP

We can easily plot the feasible region and the objective function. Then we observe from the plot that the optimal solution is

$x^* = 4$ chairs,	$y^* = 8$ tables,	profit = \$520.
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Solution to Problem 2 (a)

Formulation

Let c , s and w be the kilograms of corn, soybean meal and wheat bran in the batch. The batch weighs 100 kg, so

$$c + s + w = 100, \quad c, s, w \geq 0.$$

The protein requirement yields the constraint

$$0.09c + 0.60s + 0.20w \geq 30.$$

The fibre requirement yield the constraint

$$0.02c + 0.06s + 0.08w \leq 5.$$

The linear program is

$$\begin{array}{ll} \text{minimise} & 0.30c + 0.90s + 0.35w \\ \text{subject to} & c + s + w = 100 \quad (\text{batch size}) \\ & 0.09c + 0.60s + 0.20w \geq 30 \quad (\text{protein} \geq 30\%) \\ & 0.02c + 0.06s + 0.08w \leq 5 \quad (\text{fibre} \leq 5\%) \\ & c, s, w \geq 0. \end{array}$$

Solving the LP

There are three variables. Plotting the objective function and the feasible region in $\mathbb{R}^3$ is much harder. However, we can eliminate one variable from $c + s + w = 100$.

Writing $c = 100 - s - w$, the protein constraint becomes

$$0.09(100 - s - w) + 0.60s + 0.20w \geq 30 \iff 51s + 11w \geq 2100,$$

the fibre constraint becomes

$$0.02(100 - s - w) + 0.06s + 0.08w \leq 5 \iff 2s + 3w \leq 150,$$

and the objective function becomes

$$0.30(100 - s - w) + 0.90s + 0.35w = 30 + 0.60s + 0.05w.$$

We also need $c \geq 0$ which imposes another constraint that $s + w \leq 100$. So we obtain an equivalent LP

$$\min\{30 + 0.60s + 0.05w : 51s + 11w \geq 2100, \quad 2s + 3w \leq 150, \quad s + w \leq 100, \quad s, w \geq 0.\}$$

Again we can plot the objective function and the feasible region in $\mathbb{R}^2$, and by observing the plot we obtain that the optimal blend is

$$c^* = \frac{5000}{131} \text{ kg}, \quad s^* = \frac{4650}{131} \text{ kg}, \quad w^* = \frac{3450}{131} \text{ kg},$$

at a cost of 13785/262 per 100 kg batch.

Most real-world applications of LP involve a large number variables (sometimes over a million). Elementary methods that were explored in Problems 1 and 2 no longer work. As the course progresses, we will develop the theory behind the *simplex algorithm*, a powerful method for solving large-scale linear programs.

Solution to Problem 2 (b)

For each month $t = 1, 2, 3$, define

$$p_t = \text{number of units produced during month } t,$$

and

$$I_t = \text{inventory remaining at the end of month } t.$$

The initial inventory is fixed at

$$I_0 = 10.$$

In every month,

$$\text{ending inventory} = \text{beginning inventory} + \text{production} - \text{demand}.$$

Therefore,

$$I_t = I_{t-1} + p_t - d_t, \quad t = 1, 2, 3.$$

For the given data, these equations are

$$\begin{aligned} I_1 &= 10 + p_1 - 40, \\ I_2 &= I_1 + p_2 - 70, \\ I_3 &= I_2 + p_3 - 50. \end{aligned}$$

Additionally, it is required that $I_1, I_2 \geq 0$, $I_3 \geq 5$. This yields the following LP.

$$\begin{aligned} &\text{minimize} && 4p_1 + 6p_2 + 5p_3 + 0.5(I_1 + I_2 + I_3) \\ &\text{subject to} && I_1 = 10 + p_1 - 40, \\ & && I_2 = I_1 + p_2 - 70, \\ & && I_3 = I_2 + p_3 - 50, \\ & && I_3 \geq 5, \\ & && 0 \leq p_t \leq 60, \quad t = 1, 2, 3, \\ & && I_t \geq 0, \quad t = 1, 2, 3. \end{aligned} \tag{P}$$

Solution to Problem 3

Choosing the decision variables

The tempting choice is

$$n_j = \text{number of employees on duty on day } j.$$

This fails. With those variables the coverage requirements $n_j \geq r_j$ are trivially satisfiable, and there is no way to express the rule that each employee works five *consecutive* days: the variables record how many people are present, but not *which* people, so the link between one day and the next is lost.

The rule “five consecutive days on, two consecutive days off” means that there are only seven possible working patterns, one for each starting day. So we count *patterns*, not the number of people on each day. This suggests that we should consider the following decision variable:

$$x_d = \text{number of employees whose five-day block starts on day } d, \quad d = 0, \dots, 6 \text{ (Mon} \dots \text{Sun)}.$$

Formulation

Employees on duty on day j are exactly those who started on one of the days $j, j-1, j-2, j-3, j-4 \pmod{7}$, so

$$\begin{aligned} &\text{minimise} && \sum_{d=0}^6 x_d \\ &\text{subject to} && \sum_{k=0}^4 x_{(j-k) \bmod 7} \geq r_j, \quad j = 0, \dots, 6 \\ & && x_d \geq 0, \quad d = 0, \dots, 6. \end{aligned}$$

Missing constraint

The optimal solution is not guaranteed to be integer valued, and thus is not necessarily a valid solution. Strictly speaking, an integer constraint $x_d \in \mathbb{Z}$ should be imposed. But then we get IP, not LP. However, by rounding (up) the optimal solution of the LP, we get quite a good approximate solution to this real-world problem. This is very common in applications. There are problems where rounding does not give good approximation. For instance, binary variables that take value either 0 or 1. These occur often when a binary variable indicates a choice between option A and option B. Rounding usually does not give satisfying approximations. For such problems, IP is a better model than LP.